

Github Link: <https://github.com/RaghavPala/Group-06>

Language/Framework: Flask (Python)

To-Do:

1. Code Classes
 - a. Log into account (Henry)
 - b. Log out of account (Henry)
 - c. Generate QR code (Ani)
 - d. Scan code and log user present (Ani)
2. Test Cases (Henrique UC1&UC2 | Raghav UC3&UC4)
 - a. Input value diagram
 - i. UC 1

Input Element	Type	Value Specification	Valid	Invalid	Exceptional
NetID	Numeric / String	Min of 4 and Max of 9 Alphanumeric digits, first value must be a character (cannot be numeric)	Valid NetID with correct length, must be unique	NetID does not satisfy the value specification	Special characters, spaces, Empty string,
Password	Password	Length of 8-18 characters, must contain at least one special character and at least one number	Valid password with correct length and includes specifications , also must be unique	Password does not satisfy the value specification	Containing spaces or control characters
Login	Action	User must press the login button	User presses login button	User cancels out or does not press login button	Timeout on the login page

ii. UC 2

Input Element	Type	Value Specification	Valid	Invalid	Exceptional
Logout	Action	User must press logout button	User presses logout button	User does not press logout button / User stays on main dashboard page	

iii. UC 3

Input Element	Type	Value Specification	Valid	Invalid	Exceptional
Instructor Session	Session	User must be logged in and is_instructor = True	Logged in as instructor	Logged in as student	No active session
Course ID	String	Passed as query parameter course_id; should refer to an existing course	Existing course ID like CS3354.001	Invalid / nonexistent course ID	Missing / empty course_id
Generate Code Action	Action	User sends GET request to generate enrollment code	Request sent successfully	User does not send request	Server/database issue during request

iv. UC 4

Input Element	Type	Value Specification	Valid	Invalid	Exceptional
Instructor Session	Session	User must be logged in and is_instructor = True	Logged in as instructor	Logged in as student	No active session
Token	String	JSON body must include token; token must be properly formatted and signed	Valid token generated by system	Malformed or tampered token	Missing token
Scan Action	Action	User sends POST request to scan attendance	Request sent successfully	User does not send request	Request interrupted / server error

b. Determine test case scenarios/output

i. UC 1

Test Case #	NetID	Password	Login Action	Expected Outcome
1 (User - Student)	Valid	Valid	Valid	Redirect to Student Dashboard
2	Valid	Invalid	Valid	Error message: "Incorrect password"
3	Invalid	Valid	Valid	Error message: "Invalid NetID format"
4	Exception	Valid	Valid	Error message: "Invalid NetID format"
5	Valid	Exception	Valid	Error message: "Incorrect password"
6	Valid	Valid	Invalid / Cancel	No action / remain on login page

7 (User - Professor)	Valid	Valid	Valid	Redirect to Instructor Dashboard
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ii. UC 2

Test Case #	Logout	Expected outcome
1	Valid	Show logout successful page, navigate back to login page
2	Invalid	No action/ stays on page

iii. UC 3

Test Case #	Instructor Session	Course ID	Generate Code Action	Expected Outcome
1	Valid	Valid	Valid	Enrollment code generated successfully
2	Invalid	Valid	Valid	Error message: "access denied"
3	Exception	Valid	Valid	Error message: "not authenticated"
4	Valid	Invalid	Valid	Error message: "course not found"
5	Valid	Exception	Valid	Error message: "course not found"
6	Valid	Valid	Invalid / Cancel	No action / remain on current page

iv. UC 4

Test Case #	Instructor Session	Token	Scan Action	Expected Outcome
1	Valid	Valid	Valid	Attendance marked successfully
2	Exception	Valid	Valid	Error message: "not authenticated"
3	Invalid	Valid	Valid	Error message: "access denied"
4	Valid	Invalid	Valid	Error message: "invalid token: malformed token"
5	Valid	Exception	Valid	Error message: "missing token"
6	Valid	Valid	Invalid / Cancel	No action / remain on scan page
7	Valid	Invalid	Valid	Error message: "invalid token: invalid signature"
8	Valid	Invalid	Valid	Error message: "invalid token: token expired"
9	Valid	Valid	Valid	Error message: "no active session"
10	Valid	Valid	Valid	Error message: "already marked present"

c. Give concrete values for test cases

i. UC 1

Test Case #	NetID	Password	Login Action	Expected Outcome (based on current app.py)
1	dal123456	password1234*	Click Login	Redirect to Student Dashboard
2	dal123456	wrongpassword	Click Login	Error: Incorrect password
3	1abc	password1234*	Click Login	Error: Invalid NetID format
4	dal@123	password1234*	Click Login	Error: Invalid NetID format
5	dal123456	pass 1234*	Click Login	Error: Incorrect password
6	dal123456	password1234*	Cancel / No Click	No action / remain on login page
8	proftest	profpas1*	Click Login	Redirect to Instructor Dashboard

ii. UC 2

Test Case #	Logout	Expected outcome
1	Click Logout	Logout successful → redirected to login page
2	No Click	No action -> remain on same page

iii. UC 3

Test Case #	Instructor Session	Course ID	Generate Code Action	Expected Outcome
1	Logged in as instructor (proftest)	CS3354.001	Click / request generate code	Enrollment code generated successfully
2	Logged in as student (dal123456)	CS3354.001	Click / request generate code	Error: access denied
3	No login session	CS3354.001	Click / request generate code	Error: not authenticated
4	Logged in as instructor (proftest)	BAD999	Click / request generate code	Error: course not found
5	Logged in as instructor (proftest)	empty / missing value	Click / request generate code	Error: course not found
6	Logged in as instructor (proftest)	CS3354.001	No click / no request	No action / remain on current page

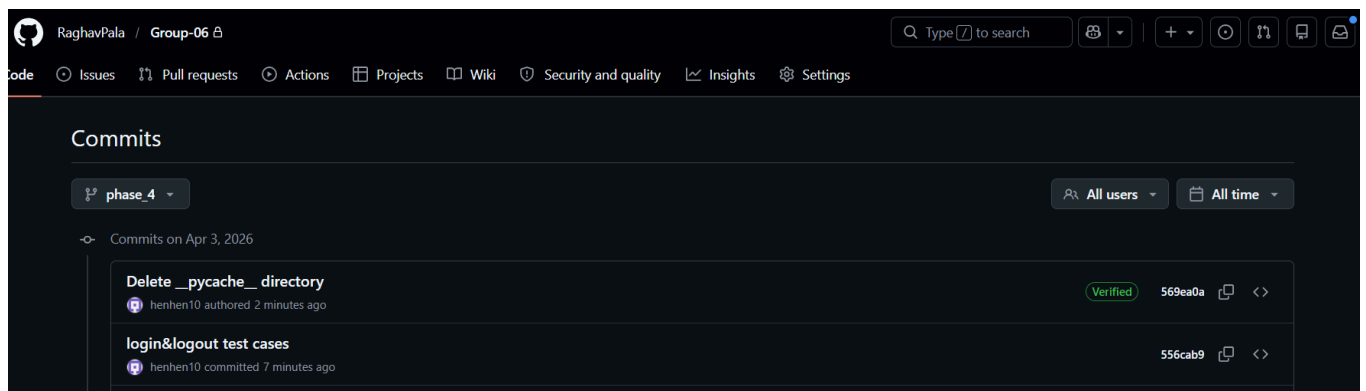
iv. UC 4

Test Case #	Instructor Session	Token	Scan Action	Expected Outcome
1	Logged in as instructor (proftest)	Valid token generated for dal123456, CS3354.001	Click / submit scan	Attendance marked successfully
2	No login session	Valid token generated for dal123456, CS3354.001	Click / submit scan	Error: not authenticated
3	Logged in as student (dal123456)	Valid token generated for dal123456, CS3354.001	Click / submit scan	Error: access denied
4	Logged in as instructor (proftest)	badtoken	Click / submit scan	Error: invalid token: malformed token
5	Logged in as instructor (proftest)	missing / empty token	Click / submit scan	Error: missing token

6	Logged in as instructor (proftest)	Valid token generated for dal123456, CS3354.001	No click / no submit	No action / remain on scan page
7	Logged in as instructor (proftest)	Tampered token with changed NetID/signature mismatch	Click / submit scan	Error: invalid token: invalid signature
8	Logged in as instructor (proftest)	Expired token from old epoch	Click / submit scan	Error: invalid token: token expired
9	Logged in as instructor (proftest)	Valid token generated for dal123456, CS3354.001	Click / submit scan	Error: no active session
10	Logged in as instructor (proftest)	Valid token generated for dal123456, CS3354.001	Click / submit scan	Error: already marked present

d. Code/implement test cases

UC1&UC2 test cases pushed to main branch of repo

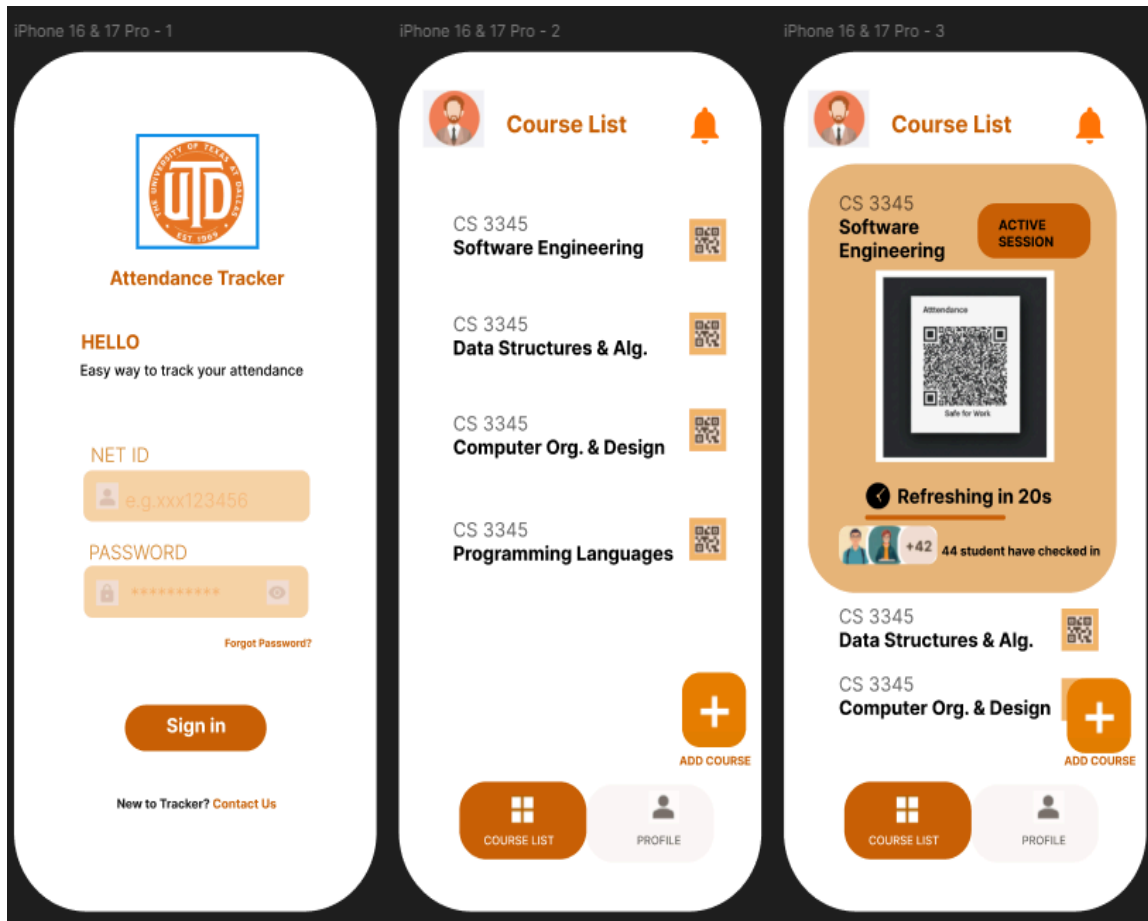


UC3&UC4 test cases pushed

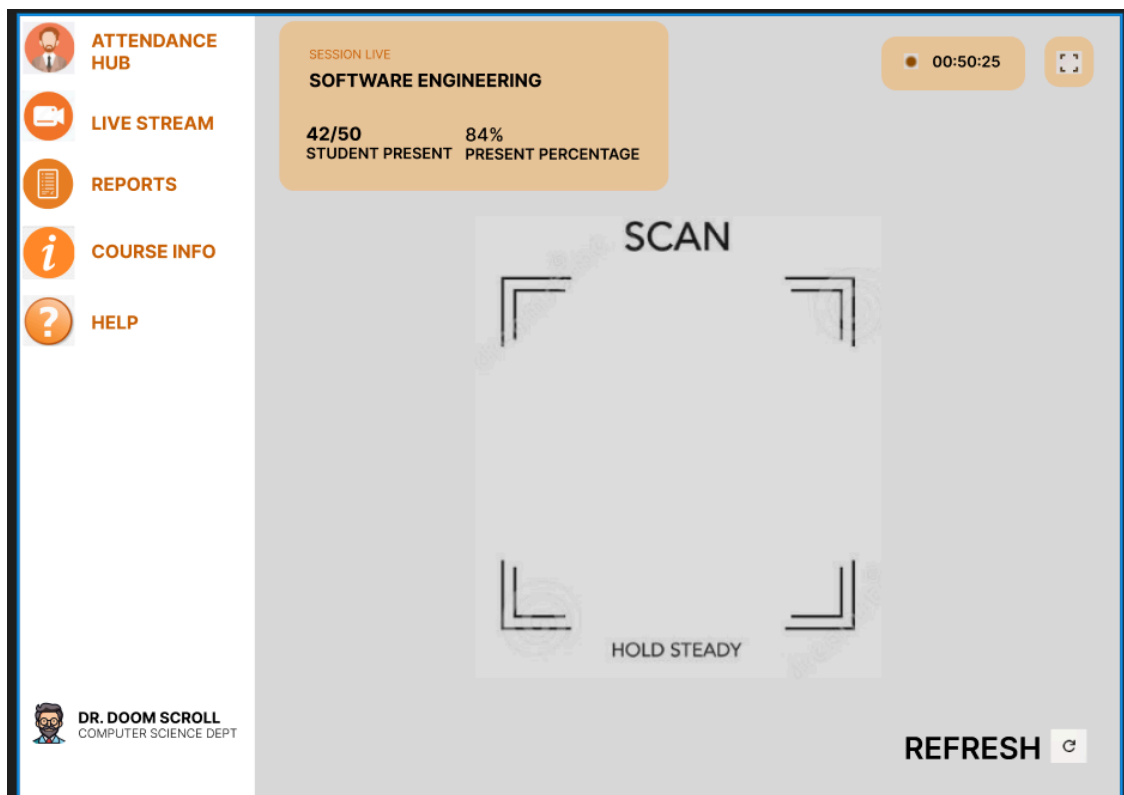
test_attendance.py	Test cases for UC3 and UC4	4 minutes ago
test_generate_code.py	Test cases for UC3 and UC4	4 minutes ago

3. Create wireframes (Trevor, Anish)

a. Use site map for specific pages needed



Wireframe for the student to check in



Wireframe for the professor to scan the students QR .

Figma share link-

<https://www.figma.com/design/mmKL2GSUWUmTk8XwJ3jlp/Untitled?node-id=0-1&t=FwveR8D5wXhX2t0h-1>